STRUCTURAL DESIGN AND FEASIBILITY ASSESSMENT OF LOCALLY SOURCED ALUMINUM AND COMPOSITE MATERIALS FOR A CUBESAT STRUCTURAL ENGINEERING MODEL

SYLVESTER OGAGOGHENE AGOSE

(210251019)

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Sylvester Ogagaoghene AGOSE
(210251019)

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CERTIFICATION

This project with the title "STRUCTURAL DESIGN AND FEASIBILITY ASSESSMENT OF LOCALLY SOURCED ALUMINUM AND COMPOSITE MATERIALS FOR A CUBESAT STRUCTURAL ENGINEERING MODEL" submitted by Sylvester Ogagaoghene AGOSE was carried out under my supervision at Lagos State University.

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Dr. R. O. Durojaye Date
Project Supervisor

TABLE OF CONTENTS

LIST OF TABLES

LIST OF FIGURES

GLOSSARY OF TECHNICAL TERMS,

LIST OF ABBREVIATIONS, AND NOMENCLATURE

SECTION I: GLOSSARY OF TECHNICAL TERMS

0.1 Standards & Specifications

CubeSat Design Specification (CDS)

Official technical document by Cal Poly defining all mechanical, dimensional, and interface requirements for CubeSat structures, including rail width tolerances (8.5 ± 0.1 mm), mass limits, center-of-gravity constraints, and surface treatment requirements for P-POD compatibility.

NASA GEVS (GSFC-STD-7000A)

NASA's General Environmental Verification Standard defining environmental test levels for flight qualification.

SMC-S-016

U.S. Air Force Space and Missile Systems Center test standard for launch, upper-stage, and space vehicles, used alongside NASA GEVS for certain mission profiles.

ASTM E8

Standard Test Method for Tension Testing of Metallic Materials. Prescribes dogbone specimen geometry, test machine settings, and procedures for determining yield strength, UTS, and elastic modulus of metals.

ASTM D638

Standard Test Method for Tensile Properties of Plastics. Defines specimen geometry (Type I–V), test speed, and reporting for polymers and composites. Type IV specimens are used in this study in XY, XZ, and ZX print orientations to quantify FDM anisotropy.

PC/104 Standard

Embedded computer bus standard widely adopted for stackable CubeSat avionics boards (OBC, EPS, communications).

0.2 Satellite Classes & Models

1U CubeSat

The fundamental CubeSat form factor: 100 × 100 × 113.5 mm, maximum mass ~1.33 kg. The '1U' baseline from which larger configurations (2U, 3U, 6U) are derived.

Structural Engineering Model (SEM)

A high-fidelity physical replica built from intended flight materials, loaded with mass dummies to simulate real mass distribution. Used exclusively for structural and vibration testing without functional avionics.

Showpiece / Display Model

A visually accurate but non-structural replica fabricated from PLA-CF for educational display. Demonstrates FDM geometric complexity without requiring structural integrity.

Mass Dummy

An inert aluminum block machined to the mass and approximate dimensions of real avionics (battery, OBC, EPS). Integrated into the SEM to accurately simulate inertial loading during vibration testing, bringing total mass to 1.33 kg.

0.3 Materials: Aluminum Alloys

Aluminum 6061-T6

Aerospace-standard CubeSat structural alloy (Al-Mg-Si). T6 temper yields ~276 MPa yield strength, 310 MPa UTS, and 68.9 GPa elastic modulus via Mg_2Si precipitation hardening. Serves as the control material in this study.

Aluminum 6063-T6

The 'architectural alloy' dominant in the Lagos market. Lower Mg/Si content limits T6 yield strength to ~214 MPa (~22% below 6061) but maintains elastic modulus at ~68.9 GPa making it viable for stiffness-critical CubeSat design.

Aluminum 6063-T5

Lower-temper variant of Al 6063, air-cooled from the extrusion die and artificially aged without prior solution heat treatment. Achieves ~145–160 MPa yield strength, the weakest commercially available Al 6063 product from Lagos vendors.

Recycled Aluminum Scrap

Secondary aluminum from Lagos foundries, processed from mixed feedstocks (beverage cans, automotive heads).

T6 Temper

Two-step heat treatment: (1) solution heat treatment at ~530°C followed by rapid quench; (2) artificial aging at ~160–175°C to precipitate fine Mg_2Si hardening phases. Produces peak mechanical properties in 6xxx series alloys.

Yield Strength (0.2% Offset)

Stress at which a material exhibits 0.2% permanent deformation. Determined from the stress-strain curve as the primary design allowable.

Ultimate Tensile Strength (UTS)

Maximum engineering stress sustained before fracture. Structures are typically sized against yield strength with safety factors, as plastic deformation is unacceptable in flight hardware.

Elastic Modulus (Young's Modulus, E)

Material stiffness; the stress-to-strain ratio in the elastic regime. The critical property for stiffness-driven CubeSat design, governing natural frequency.

Specific Strength

Yield or ultimate tensile strength normalized by material density (units: kN·m/kg or $MPa \cdot cm^3/g$). A mass-efficiency metric enabling fair comparison between materials of different densities.

Specific Modulus

Elastic modulus normalized by material density (units: $GPa \cdot cm^3/g$ or MN·m/kg). Determines stiffness-per-unit-mass, governing the natural frequency of a structure at a given mass.

Stiffness-to-Mass Ratio

A structural-level metric (distinct from specific modulus) relating the global stiffness of an assembled structure to its total mass. Directly determines the fundamental natural frequency: $f_{\square} \propto \sqrt{(k/m)}$.

0.4 Materials: Composites & Polymers

Carbon Fiber Reinforced Polymer (CFRP)

Composite material with carbon fiber reinforcement in a polymer matrix. Short (chopped) carbon fibers at 15–20 wt% are dispersed in Nylon 12 or PLA for FDM processing. Density ≈ 1.2–1.4 g/cm^3 vs aluminum at 2.7 g/cm^3, enabling significant mass reduction.

Nylon-CF (PA12-CF)

Polyamide 12 reinforced with ~15–20 wt% chopped carbon fibers. Primary composite material for the SEM chassis. XY tensile strength: 76–84 MPa. Hygroscopic — requires pre-drying at ~70°C for 4–8 hours before printing to prevent porosity and delamination.

PLA-CF (Carbon Fiber PLA)

Polylactic Acid reinforced with chopped carbon fibers. Used for the Showpiece model due to lower print temperature (~190–220°C) and dimensional stability. Mechanically inferior to PA12-CF at elevated temperatures but sufficient for ground-based display.

PEEK (Polyether Ether Ketone)

Industry-standard space polymer offering continuous use to ~250°C, near-zero outgassing, and high specific strength. Requires printers with heated chambers (~120°C) and nozzle temperatures ~380°C (cost > $100,000) — inaccessible in the Lagos context.

Outgassing
Release of volatile compounds from materials in vacuum. Can contaminate optical payloads or contribute to cold welding.

Coefficient of Thermal Expansion (CTE)
Fractional dimensional change per degree temperature change (μm/m·K). CTE mismatch in LEO thermal cycling (−40°C to +80°C) generates significant thermal stresses in rigidly joined metal-polymer assemblies, requiring compliant joint designs.

0.5 Manufacturing Processes

Fused Deposition Modeling (FDM)
Additive manufacturing process in which thermoplastic filament is heated, extruded through a nozzle, and deposited layer-by-layer. The layer-by-layer deposition physics introduces directional mechanical properties (anisotropy). Also termed Fused Filament Fabrication (FFF).

Anisotropy (FDM)
Direction-dependent mechanical properties in FDM parts.

Print Orientation (XY, XZ, ZX)
Angular relationship between a part's load axis and the FDM build direction. XY (Flat): highest strength, fibers aligned in load direction. XZ (On-Edge): intermediate. ZX (Upright): lowest strength, layer bonds loaded in tension. ASTM D638 specimens are printed in all three orientations to quantify the anisotropy ratio.

Hard Anodization (Type III)
Electrochemical surface treatment producing a 50–100 μm Al_2O_3 layer on aluminum. Industry-standard CubeSat rail treatment per CDS, providing exceptional hardness and preventing P-POD galling and cold welding.

Electroless Nickel Plating
Autocatalytic chemical deposition of a nickel-phosphorus alloy coating. Available from Lagos marine/automotive finishing shops. Provides hardness and galling resistance comparable to hard anodization; a viable local alternative for CubeSat rail surface treatment.

Heat-Set Threaded Insert
Brass nut with a knurled exterior installed by thermal press-fit into FDM-printed thermoplastic. Heated to ~250°C, the surrounding polymer reflows into the knurls, creating a composite interface.

Design for Manufacturing (DFM)
Engineering methodology that optimizes design to match available manufacturing capabilities.

CNC Machining

Subtractive manufacturing using computer-controlled cutting tools to produce precision geometry from aluminum billets. Used to machine the CubeSat rails to the CDS-required tolerance of 8.5 ± 0.1 mm rail width.

0.6 Structural Engineering & Analysis

Finite Element Analysis (FEA)

Computational method discretizing a continuous body into finite elements to solve structural mechanics equations. Used in ANSYS/SolidWorks Simulation to evaluate the hybrid chassis under quasi-static (15g) and random vibration loads, predicting modal frequencies, stress distributions, and safety margins.

Boundary Conditions (FEA)

Constraints applied to the FE model that define how the structure is supported and loaded, replicating the physical deployment scenario.

Von Mises Stress

Scalar equivalent stress combining all stress components via the distortion energy criterion. Failure predicted when Von Mises stress exceeds uniaxial yield strength.

Safety Factor (SF)

Ratio of material yield strength to applied stress.

A-Basis vs B-Basis Allowables

Statistical design allowables derived from material test data. A-Basis: the 5th percentile value (95% of specimens meet or exceed); used for single-load-path (critical) structures. B-Basis: the 10th percentile value (90% probability); used for redundant load-path structures.

Stress Concentration Factor (K□)

Dimensionless multiplier (K□ ≥ 1) quantifying the local stress amplification at geometric discontinuities; holes, fillets, notches, and thickness changes.

Damping Ratio (ζ)

Dimensionless parameter ($0 \leq \zeta \leq 1$) characterizing a structure's rate of free-vibration energy dissipation, normalized by critical damping: $\zeta = c / (2\sqrt{(km)})$.

Natural Frequency (f□)

The frequency of free oscillation of a structure, determined by stiffness-to-mass ratio: f□ = $(1/2\pi)\sqrt{(k/m)}$.

Modal Analysis

Technique to identify natural frequencies, mode shapes, and damping ratios of a structure. Performed computationally via FEA eigenvalue solution or experimentally via sine sweep/impact hammer testing.

Random Vibration

Broadband stochastic vibration environment (20–2000 Hz) during launch ascent, characterized by a Power Spectral Density (PSD) curve (G^2/Hz).

Power Spectral Density (PSD)

Frequency-domain statistical description of random vibration: mean-square acceleration per unit frequency (G^2/Hz). Area under the PSD curve equals mean-square acceleration; its square root is the Gr^{ML} value used to define test inputs.

Quasi-Static Load

Slowly varying acceleration is treated as a static force for structural analysis.

Delamination

Layer separation in FDM composites when Z-axis tensile stress exceeds interlayer bond strength (~25–40 MPa for PA12-CF).

3σ Stress

Peak stress at three standard deviations above the mean in the Gaussian stress distribution under random vibration (99.73% probability envelope).

Margin of Safety (MoS)

Structural reserve beyond the required safety factor: MoS = (Allowable / (Applied × SF)) − 1.

0.7 Space Mechanisms & Hardware

P-POD (Poly-Picosatellite Orbital Deployer)

Standard CubeSat deployment mechanism (Cal Poly). Houses up to three 1U satellites in a spring-loaded, rail-guided tube.

Cold Welding

Adhesion between unprotected metal surfaces in vacuum via atomic bonding at clean metal-to-metal interfaces, without elevated temperature.

COTS (Commercial Off-The-Shelf)

Pre-manufactured standard components are procured rather than custom-developed.

LEO (Low Earth Orbit)

Orbital altitudes ~200–2,000 km.

Galling

Severe adhesive wear at sliding metal interfaces under pressure, causing material transfer and progressive seizing.

Thermal Cycling

Repeated temperature oscillation in LEO (−40°C to +80°C per 90-minute orbit; ~5,800 cycles/year).

Outgassing (in situ)

See Outgassing under Composites & Polymers.

0.8 Engineering Concepts & Frameworks

Frugal Engineering

Design philosophy maximizing functional utility per unit cost; replacing expensive materials with rigorous characterization and intelligent geometry. 'Good enough' is valid only when verified by testing. Not about accepting lower safety, but eliminating unnecessary performance margins that add cost without adding function.

Material-Standard Disconnect

The central problem of this research: a systemic misalignment between aerospace standard material specifications (which assume traceable aerospace supply chains) and the industrial realities of developing economies.

Hybrid Skeleton Architecture

The design philosophy adopted for this chassis: aluminum four-rail primary structure carries compressive launch loads; FDM Carbon Fiber Nylon panels provide lateral stiffness, shear resistance, and PCB mounting. Exploits each material's dominant advantage while avoiding its weakness.

Material Atlas

A documented dataset of measured mechanical properties (yield strength, UTS, E, and statistical variance) for locally sourced Lagos materials.

Traceability (Materials)

The ability to track a material's complete history from smelting through heat treatment and machining, with certification at each step. Required by global aerospace standards for flight hardware.

Stiffness-Driven Design

Structural design approach where the primary constraint is achieving minimum stiffness to maintain natural frequency above the launch vehicle's forcing frequencies, rather than meeting a minimum strength threshold.

Coefficient of Variation (CoV)

Normalized statistical dispersion: $CoV = (\sigma / \mu) \times 100\%$.

0.9 Testing Methods

Sine Sweep Testing

Vibration test method sweeping sinusoidal excitation across 20–2000 Hz. Resonance peaks in the structural response identify natural frequencies.

Impulse Response Testing

Experimental modal analysis using an instrumented impact hammer to deliver a broadband impulse. Accelerometer response is transformed via FFT to extract natural frequencies and damping ratios.

Fast Fourier Transform (FFT)

An algorithm converting a time-domain vibration signal to a frequency-domain amplitude spectrum. Used in impulse response testing to identify natural frequencies from the free-vibration decay signal.

Static Load Testing

Tertiary fallback method applying calibrated dead weights equivalent to 15g × 1.33 kg ≈ 196 N. Dial indicators verify elastic recovery (no permanent deformation). Validates static strength but not dynamic performance or fatigue resistance.

Universal Testing Machine (UTM)

Precision load frame applying controlled tensile loads at specified strain rates. Used per ASTM E8 and D638 for Phase I material characterization.

Dogbone Specimen

Tensile specimen with wide grip ends tapering to a narrow uniform gauge section, concentrating failure within the gauge region. Geometry specified by ASTM E8 (metals) and ASTM D638 (plastics). Aluminum dogbones are CNC-machined; composite specimens are FDM-printed.

1.0 Facilities & Organizations

Unilag Fabrication Lab (FabLab)

Fabrication laboratory at the University of Lagos housing CNC machining and FDM printing equipment, including printers with hardened steel nozzles capable of processing abrasive carbon fiber filaments. Primary manufacturing facility for this project.

NASRDA

National Space Research and Development Agency; Nigeria's civil space agency responsible for the NigeriaSat satellite program and space technology development.

Standards Organisation of Nigeria (SON)

Nigeria's national standards body. Its laboratory is identified as a potential venue for Universal Testing Machine access during the Phase I ASTM tensile testing program.

SECTION II: LIST OF ABBREVIATIONS

The following abbreviations are used throughout this thesis. All abbreviations are defined at their first occurrence in the text.

Abbrev.	**Full Form**	**Abbrev.**	**Full Form**
AM	Additive Manufacturing	**ASTM**	American Society for Testing and Materials
CAD	Computer-Aided Design	**CAM**	Computer-Aided Manufacturing
CDS	CubeSat Design Specification	**CF**	Carbon Fiber
CFRP	Carbon Fiber Reinforced Polymer	**CNC**	Computer Numerical Control
CoV	Coefficient of Variation	**COTS**	Commercial Off-The-Shelf
CTE	Coefficient of Thermal Expansion	**CVCM**	Collected Volatile Condensable Material
DFM	Design for Manufacturing	**EPS**	Electrical Power System
FDM	Fused Deposition Modeling	**FEA**	Finite Element Analysis
FFF	Fused Filament Fabrication	**FFT**	Fast Fourier Transform
GEVS	General Environmental Verification Standard	**GPS**	Global Positioning System
GSFC	Goddard Space Flight Center	**LEO**	Low Earth Orbit
MoS	Margin of Safety	**NASRDA**	National Space Research and Development Agency (Nigeria)
NASA	National Aeronautics and Space Administration	**OBC**	On-Board Computer
PA12	Polyamide 12 (Nylon 12)	**PA12-CF**	Polyamide 12 Carbon Fiber Composite
PCB	Printed Circuit Board	**PEEK**	Polyether Ether Ketone
PEI	Polyetherimide	**PLA**	Polylactic Acid
PLA-CF	Polylactic Acid Carbon Fiber Composite	**P-POD**	Poly-Picosatellite Orbital Deployer
PSD	Power Spectral Density	**SEM**	Structural Engineering Model
SF	Safety Factor	**SMC**	Space and Missile Systems Center
SON	Standards Organisation of Nigeria	**SSTL**	Surrey Satellite Technology Limited
TML	Total Mass Loss	**TVAC**	Thermal Vacuum

Abbrev.	**Full Form**	**Abbrev.**	**Full Form**
UTS	Ultimate Tensile Strength	**UTM**	Universal Testing Machine
wt%	Weight Percent	**1U**	One Unit (CubeSat form factor)

SECTION III: NOMENCLATURE (SYMBOLS)

Symbols used in equations, figures, and tables throughout this thesis are defined as follows. Subscripts and superscripts are defined alongside the parent symbol where applicable.

Symbol	Description	Units
E	Elastic Modulus (Young's Modulus)	GPa or MPa
f□	Natural Frequency	Hz
f_1	First (fundamental) Natural Frequency	Hz
Gr^{ML}	Root-Mean-Square Acceleration	g
I	Second Moment of Area	mm^4
k	Structural Stiffness	N/m
K	Effective Length Factor (Buckling)	dimensionless
K□	Stress Concentration Factor	dimensionless
$[K]$	Global Stiffness Matrix (FEA)	N/m
L	Unsupported Column Length	mm or m
m	Mass	kg
$[M]$	Global Mass Matrix (FEA)	kg
MoS	Margin of Safety	dimensionless
n	Sample Size	dimensionless
P^{KR}	Euler Buckling Load	N
PSD	Power Spectral Density	G^2/Hz
σ	Normal Stress	MPa
σ_γ	Von Mises Equivalent Stress	MPa
$\sigma_\gamma{}^c$	Yield Strength (0.2% Offset)	MPa
$\sigma_{u}n_{i}$	Ultimate Tensile Strength	MPa
σ□$_a{}^x$	Maximum Local Stress (at discontinuity)	MPa
σ□$_o$□	Nominal (average) Stress	MPa
$\sigma_3\sigma$	Three-Sigma Peak Stress (Random Vibration)	MPa
SF	Safety Factor	dimensionless
α	Coefficient of Thermal Expansion (CTE)	µm/m·K
ε	Strain	mm/mm (dimensionless)
ω	Angular Natural Frequency	rad/s

Symbol	Description	Units
$\boldsymbol{\omega_n}$	nth Natural Frequency (Eigenvalue)	rad/s
$\boldsymbol{\zeta}$	Damping Ratio	dimensionless
$\boldsymbol{\{\varphi\}}$	Mode Shape Vector (Eigenvector)	dimensionless
$\boldsymbol{\mu}$	Mean Value	varies
$\boldsymbol{\sigma_{data}}$	Standard Deviation of Test Data	varies
$\boldsymbol{\Delta T}$	Temperature Differential	°C or K
$\boldsymbol{\Delta L}$	Thermal Elongation	mm
$\boldsymbol{\rho}$	Material Density	g/cm^3 or kg/m^3

ABSTRACT

The global CubeSat Design Specification (CDS) and NASA's General Environmental Verification Standard (GEVS) implicitly assume access to aerospace-grade Aluminum 6061-T6, a material that is not readily available within Nigeria's domestic supply chain due to high importation costs and logistical constraints. This study investigates the structural feasibility of a hybrid 1U CubeSat chassis that replaces 6061-T6 with locally sourced Aluminum 6063 and recycled aluminum from Lagos foundries, complemented by additively manufactured Carbon Fiber–reinforced Nylon (PA12-CF) components.

A five-phase methodology was adopted: (1) physical tensile characterization of candidate materials in accordance with ASTM E8 and D638 standards, (2) load-path-optimized structural design, (3) Finite Element Analysis under quasi-static (15g) and random vibration (20–2000 Hz) loading conditions, (4) fabrication of a 1U Structural Engineering Model (SEM), and (5) experimental modal validation.

Although Aluminum 6063 exhibits approximately 22% lower yield strength than 6061-T6, it is hypothesised that the inherent structural margins of the 1U form factor permit its safe application in primary load-bearing rails when appropriate geometric reinforcement and print-orientation strategies are implemented to mitigate FDM anisotropy. Structural viability is defined by a first natural frequency above 100 Hz, von Mises stresses below 50% of material yield strength, and full CDS dimensional compliance.

This work advances a practical framework for frugal, indigenous satellite development, one that is replicable across resource-constrained institutions in sub-Saharan Africa and other emerging spacefaring nations.

CHAPTER ONE

INTRODUCTION

1.1 BACKGROUND AND CONTEXT

The CubeSat specification, developed in 1999 through the collaboration of California Polytechnic State University (Cal Poly) and Stanford University, marked a significant revolution in the aerospace industry. It established a standardized 10 × 10 × 10 cm dimension, called "1U," that weighs about 1.33 kg *(Napoli, 2013; Rawls, 2014; The CubeSat Program, Cal Poly SLO, 2022)*. With the establishment of the CubeSat approach, space explorations are no longer burdened by the high cost of launching a ton or multi-ton payloads to LEO.

The general specification approved for the fabrication of the CubeSat structure involves Aerospace Grade Aluminum Al6061-T6/Al7075-T6 precipitation-hardened alloys *(The CubeSat Program, Cal Poly SLO, 2022)***.** These are chosen based on their high yield strength, ease of machinability, and low outgassing rates, which are critical to prevent cold welding in the Poly-Picosatellite Orbital Deployer referred to as P-POD *(Napoli, 2013)*.

In the case of Nigeria, there is no local supply chain for such aerospace-grade materials or it would be cost-prohibitive due to importation costs, tariffs, and lead times. Moreover, the Lagos market in particular is dominated by Al 6063, primarily used for architectural extrusions such as windows and roofs, in addition to a large unregulated supply of recycled aluminum scrap materials processed in domestic foundries. This creates a supply chain mismatch between the requirements for such materials in accordance with the global CubeSat Design Specification and what is practical in the local Nigerian market. On the other hand, the evolution of Additive Manufacturing (AM) technology, specifically in Fused Deposition Modeling (FDM) with composite materials, offers a conceivable solution. The addition of short-carbon fiber (CF) with a Thermoplastic matrix of Nylon (Polyamide 12) or polylactic acid (PLA) provides a material with equivalent stiffness properties to those of aluminum *(Ramírez-Prieto et al., 2025)*. This research aims for a paradigm shift with a Hybrid Structural Engineering Model that leverages the accessibility of locally sourced aluminum for the use in main load-carrying rails, augmented with the addition of 3D-printed carbon fiber-reinforced parts for the internal structure. The goal is to prove the feasibility of "Frugal Engineering," which can support the tight specifications of launch services companies regarding vibrations and acoustic performance, while also rooting the industry supply chain within the locale

1.2 MOTIVATION OF THE STUDY

The reliance on imported raw materials for technological projects in Nigeria introduces severe vulnerabilities, including exposure to foreign exchange volatility, customs delays, and supply chain disruptions. As noted in economic analyses of local vs. imported sourcing, the "true cost" of importation includes not just the purchase price but logistics and time *(Obomeghie & Obomeghie, 2025)*. For a student or university-led CubeSat project, a 3-month delay to import a specific aluminum block can derail an entire academic timeline.

Local vendors in Lagos supply aluminum products primarily for the construction and marine sectors. These are typically Al 6063 (architectural alloy) or Al 5052 (marine grade). The unavailability of 6061 is driven by market demand; there is little industrial call for the high-strength, heat-treatable alloys required by aerospace standards. Furthermore, the informal recycling sector produces vast quantities of secondary aluminum. While "green" and cost-effective, this material is often cast into pot-grade aluminum with little quality control *(Meagher, 2017)*. Utilizing this material for a high-reliability application like a satellite structure requires a rigorous characterization and selection process, which this project addresses.

Lagos has seen a growing "Maker Movement" with the establishment of hubs like the Unilag Fabrication Lab, which possesses 3D printing and CNC machining capabilities. The availability of FDM printers capable of handling abrasive filaments (using hardened steel nozzles) opens the door to using Carbon Fiber reinforced plastics (CFRP) *(Bochnia et al., 2021; Islam et al., 2025; Ramírez-Prieto et al., 2025).*

1.3 PROBLEM STATEMENT

The central problem driving this research is the "Material-Standard Disconnect" in the development of indigenous space hardware in emerging economies.

The global aerospace standards that govern CubeSat acceptance, specifically the *Cal Poly CubeSat Design Specification (CDS)* and launch environment standards like *NASA GEVS (GSFC-STD-7000)*, are predicated on the use of high-fidelity, traceable aerospace materials (Al 6061-T6, Al 7075, Titanium). These standards implicitly assume a supply chain capable of delivering materials with guaranteed yield strengths (>270 MPa) and specific fatigue limits *(Goddard Space Flight Center, 2018; Rawls, 2014; The CubeSat Program, Cal Poly SLO, 2022).*

In the Nigerian context, adhering strictly to these material specifications necessitates the importation of raw stock, which is economically inefficient and logistically fragile. The locally available alternatives, Aluminum 6063 and recycled scrap, are chemically distinct and mechanically inferior. Al 6063 has a lower yield strength and different hardening response, while recycled scrap suffers from porosity and brittle intermetallic inclusions due to uncontrolled iron content *(Gao et al., 2026; Nunes et al., 2023)*.

Furthermore, FDM carbon fiber composites introduce anisotropic failure risks under launch vibration loads that require deliberate design mitigation

There is currently no established design methodology or qualification dataset for a Hybrid CubeSat Chassis that successfully integrates Nigerian-sourced Aluminum 6063/Scrap with FDM Carbon Fiber Composites. Consequently, indigenous developers are forced to choose between the high cost of compliance (importation) or the high risk of failure (using unverified local materials). This research seeks to bridge this gap by rigorously assessing the feasibility of such a hybrid system.

1.4 AIM AND OBJECTIVES

1.4.1 AIM OF RESEARCH

The primary aim of this project is to design, fabricate, and structurally validate a **Structural Engineering Model (SEM)** of a 1U CubeSat chassis that utilizes a hybrid construction of locally sourced aluminum (from the Lagos market) and additive-manufactured carbon fiber reinforced polymers, ensuring compliance with standard launch vehicle vibration requirements.

This SEM validates structural design feasibility and material performance. Full qualification (requiring complete environmental testing per GEVS, including thermal-vacuum and flight-traceable materials) is recommended as future work, but is outside the scope of this undergraduate project.

1.4.2 OBJECTIVES OF RESEARCH

To achieve the aim, the following specific objectives are defined:

1. **Material Characterization:** To experimentally determine the tensile strength, yield strength, and Elastic Modulus of locally sourced Aluminum 6063 and recycled aluminum scrap using **ASTM E8** standards, and to characterize the anisotropic properties of FDM Carbon Fiber PLA/Nylon using **ASTM D638** *(ASTM International, 2022)*.
2. **Design Development:** To design a hybrid 1U CubeSat structure that strategically allocates loads, utilizing aluminum for the primary rails (longitudinal compression) and 3D-printed composites for the secondary frame, incorporating design-for-manufacturing (DFM) principles for local CNC and FDM capabilities.
3. **Computational Simulation:** To perform Finite Element Analysis (FEA) to simulate the structural response under Quasi-Static loads (15g) and Random Vibration (NASA GEVS levels), identifying modal frequencies, stress concentrations, and margins of safety.
4. **Fabrication of SEM:** To manufacture a high-fidelity Engineering Model using locally machined aluminum rails and FDM-printed Carbon Fiber Nylon components, ensuring strict dimensional adherence to the *CubeSat Design Specification*.
5. **Feasibility Assessment:** To validate the design through physical testing (or high-fidelity simulation proxy if shaker access is limited), verifying that the structure withstands the General Environmental Verification Standard (GEVS) acceptance levels without plastic deformation or fracture.

1.5 RESEARCH QUESTIONS

This study addresses the following critical engineering questions:

1. **Material Feasibility:** Can locally sourced Aluminum 6063 or recycled aluminum scrap from Lagos foundries provide sufficient structural margins (Yield Strength > Launch Loads × Safety Factor) to serve as the primary load-bearing rails for a 1U CubeSat?
2. **Hybrid Structural Integrity:** How does the integration of FDM-printed Carbon Fiber components affect the fundamental natural frequency of the satellite? Can the hybrid structure maintain a first mode above the 90–100 Hz stiffness requirement to avoid resonance coupling with the launch vehicle *(Goddard Space Flight Center, 2018)*?
3. **Anisotropy Mitigation:** Can specific design geometries and print orientations effectively mitigate the Z-axis weakness of FDM parts, preventing delamination during random vibration excitation *(Ramírez-Prieto et al., 2025)*?
4. **Joining Efficacy:** How do heat-set threaded inserts compare to direct threading in carbon fiber-filled filaments in terms of torque retention and pull-out strength under vibration loads *(Hogade et al., 2025)*?

1.6 SIGNIFICANCE OF THE STUDY

This research holds significant value for the advancement of space technology in developing economies:

1. **Economic Sovereignty:** By validating the use of Aluminum 6063, which costs approximately ₦2.3 million per metric ton in the local market *(WigmoreWholesale, 2026)*, versus the significantly higher landed cost of imported aerospace alloys, the structural bus cost can be reduced, improving economic viability for university space programs. This promotes the sustainability of university space programs.

2. **Frugal Innovation in Aerospace:** The study contributes to the global body of knowledge on "Frugal Engineering" for space applications. While agencies like NASA explore high-performance composites (PEEK, Titanium), this research validates a "Good Enough" approach for short-duration LEO missions, expanding the envelope of what is possible with constrained resources *(Georgantzinos, S. K., et al., 2025)*.
3. **Local Capacity Building:** By utilizing the Unilag Fabrication Lab and local foundries, the project strengthens the domestic supply chain. It provides a blueprint for other African universities to leverage local infrastructure for high-technology applications.
4. **Educational Impact:** The production of the 3D-printed showpiece and the QM provides tangible assets for teaching structural mechanics, vibration analysis, and space systems engineering, bridging the gap between theory and practice.

1.7. SCOPE AND LIMITATIONS

1.7.1 SCOPE

1. **Satellite Class:** 1U CubeSat (100 × 100 × 113.5 mm).
2. **Model Fidelity:**
    a. **Structural Engineering Model (SEM):** A functional structure made of the intended hybrid materials (Local Al + Nylon-CF), loaded with "mass dummies" to simulate electronics/batteries. This is for structural testing.
    b. **Showpiece:** A visual model made of PLA-CF for educational display.
3. **Materials:** Metals: Al 6063 (Extrusion market) and Al Scrap (Local Foundry).
    a. Composites: Carbon Fiber reinforced Nylon (PA12-CF) for the QM; Carbon Fiber PLA for the showpiece.

4. **Analysis:** Structural analysis only (Tensile properties, Modal analysis, Random Vibration response).
5. **Context:** Sourcing and fabrication limited to Lagos, Nigeria.

1.7.2 LIMITATIONS

1. **Thermal Vacuum Testing (TVAC):** The study excludes TVAC testing due to the unavailability of local facilities. The feasibility assessment is strictly structural.
2. **Radiation Analysis:** The shielding effectiveness of the hybrid structure against Lower Earth Orbit radiation (Single Event Upsets) is not evaluated.
3. **Material Variance:** The study acknowledges that recycled aluminum scrap exhibits high batch-to-batch variance. The results will be valid for the specific batches tested, with statistical safety factors recommended for generalization.
4. **Flight Heritage:** This is a ground-based feasibility study. No orbital flight demonstration is included.

CHAPTER TWO

LITERATURE REVIEW

This literature review provides a comprehensive analysis of the feasibility of a "Hybrid Structural Engineering Model" that challenges this orthodoxy. It investigates the structural viability of replacing imported aerospace alloys with locally sourced Aluminum 6063 (architectural grade) and recycled aluminum scrap processed in Lagos foundries, augmented by Additive Manufacturing (AM) using Carbon Fiber reinforced thermoplastics. The review is structured around six critical arguments: the disconnect between global standards and local supply chains; the metallurgical sufficiency of lower-grade aluminum for nanosatellite loads; the constraint of anisotropy in Fused Deposition Modeling (FDM); the engineering of hybrid joints to mitigate thermal expansion mismatch; the research gap regarding low-cost hybrid integration; and the validity of frugal innovation when backed by rigorous characterization.

The first critical argument posits that the reliance on imported aerospace-grade materials constitutes a systemic vulnerability for indigenous space programs in developing economies, necessitating a re-evaluation of local material suitability.

2.1 THE GLOBAL STANDARD AND LOCAL REALITY

2.1.1 THE GLOBAL STANDARD

The CubeSat Design Specification (CDS) explicitly recommends Aluminum 6061-T6, 7075-T6, or Titanium alloys for the primary structure. These materials are selected based on rigorous aerospace heritage. The T6 temper (solution heat-treated and artificially aged) provides a yield strength typically exceeding 270 MPa for Al 6061 and 500 MPa for Al 7075. Beyond static strength, these alloys offer high fatigue resistance, predictable thermal conductivity for heat dissipation, and low outgassing rates (Total Mass Loss < 1.0%) to prevent contamination of optical payloads.

Global standards implicitly assume a supply chain capable of "traceability", the ability to track a material's history from the smelter to the machine shop, ensuring consistent chemistry and thermomechanical processing. In the Nigerian context, however, this represents a logistical bottleneck characterized by high tariffs (up to 20% on imported metals), shipping delays of up to three months, and significant markups by local distributors for "specialty" items *(Obomeghie & Obomeghie, 2025)*.

2.1.2 THE LOCAL REALITY

Conversely, the aluminum market in Lagos is dominated by Aluminum 6063. Often termed the "architectural alloy," Al 6063 is optimized for extrudability and surface finish rather than ultimate strength. While chemically similar to 6061 (both being Al-Mg-Si alloys), 6063 contains reduced concentrations of magnesium and silicon, limiting its precipitation hardening response *(Georgantzia et al., 2021)*.

Parallel to the extrusion market is the informal recycling sector. Foundries in the Owode Onirin axis of Lagos process vast quantities of aluminum scrap, ranging from beverage cans (typically 3xxx series Mn-alloys) to automotive cylinder heads (Al-Si casting alloys). This "green aluminum" is economically accessible and supports a circular economy, but suffers from uncontrolled impurity levels. Iron contamination in secondary aluminum alloys precipitates as brittle β-Fe (Al_3FeSi) platelets, acting as stress concentrators that significantly reduce ductility and fatigue life *(Viscusi et al., 2023)*. Iron has low solubility in solid aluminum and precipitates as brittle platelets, which act as stress concentrators and significantly reduce ductility and fatigue life.

2.1.3 ECONOMIC AND LOGISTICAL COMPARISON

Imported aerospace-grade Al 6061-T6 can cost between $3.50 and $20.00 per kg, depending on the form factor and supplier markups for tolerance guarantees. In contrast, local Al 6063 extrusions are available for approximately 2.3 million Naira per metric ton (~$2.50/kg) *(Obomeghie & Obomeghie, 2025)*. Recycled scrap billets can be procured for even less, approximately $1.50/kg, though this comes with the added cost of machining and heat treatment.

The "supply chain gap" is therefore not merely a lack of material, but a misalignment between the *types* of material demanded by global standards and those supplied by the local industrial ecosystem. Bridging this gap requires verifying whether the "inferior" local materials are structurally sufficient.

TABLE 2.1: Economic and Logistical Comparison of Aluminum Supply Options for CubeSat Structural Development in Nigeria

Material	Source	Price per kg (USD)	Lead Time	Traceability	Supply Reliability
Al 6061-T6	Imported (International Distributor)	\$3.50 – \$20.00	6 – 12 weeks	Full material certification available	Low; subject to FX volatility, customs delays, and shipping disruptions
Al 6063-T6	Lagos Architectural Market (Local Extrusion Vendors)	~\$2.50 (≈ ₦2.3M/MT)	1 – 3 days	None; no material certification or batch traceability	High; widely stocked; immediate availability
Recycled Aluminum Scrap	Lagos Foundries (Owode Onirin Informal Sector)	~\$1.50	1 – 2 days	None; uncontrolled impurity levels; high batch variance	High; abundant supply, low cost, quality inconsistent

Note: Price estimates are indicative as of 2024–2025. Import prices vary significantly with order quantity, form factor (bar, sheet, extrusion), and supplier. FX conversion based on approximate rate of ₦1,600/USD. Lead times for imported material include international shipping and Nigerian customs clearance.

2.2 METALLURGICAL MECHANISMS

The second argument challenges the technical necessity of the "gold standard" Al 6061-T6 by rigorously comparing the metallurgical properties of local alternatives against the actual structural margins required for a 1U CubeSat.

The strength of 6xxx series alloys is derived from the precipitation of the metastable precursor phases of magnesium silicide (Mg_2Si). The hardening sequence typically proceeds as: Supersaturated Solid Solution $\rightarrow$ GP Zones $\rightarrow \beta'' \rightarrow \beta' \rightarrow \beta$ (stable).

1. **Aluminum 6061:** Formulated with 0.8-1.2% Magnesium and 0.4-0.8% Silicon, plus 0.15-0.40% Copper. The copper content provides additional solid-solution strengthening and facilitates a denser precipitation network, yielding a typical T6 yield strength of 276 MPa and Ultimate Tensile Strength (UTS) of 310 MPa.
2. **Aluminum 6063:** Formulated with lower solute limits (0.45-0.90% Mg, 0.20-0.60% Si) and minimal copper. This chemistry is optimized for the high extrusion speeds required by the construction industry. While corrosion resistance is superior, the lower volume fraction of Mg_2Si precipitates limits the T6 yield strength to approximately 214 MPa.

2.2.1 VIABILITY OF RECYCLED SCRAP

Recycled aluminum from Lagos foundries presents a high-variance material stream. Heat treatment studies on recycled 6xxx series alloys show that solution treatment followed by artificial ageing can partially recover mechanical properties degraded by impurity accumulation, though yield strengths for unrefined scrap remain in the range of 55–120 MPa due to residual porosity and Fe-rich intermetallics (*De Caro et al., 2023)*. However, yield strength values for unreinforced recycled scrap can be as low as 55-120 MPa if porosity and impurities are not controlled. The presence of iron-rich intermetallics makes the material brittle, reducing elongation to break to under 5%. While this lack of ductility is a concern for impact loading, satellite structures are primarily dimensioned for stiffness (Young's Modulus) to maintain natural frequencies above launch vehicle forcing functions. Since the elastic modulus of aluminum is relatively insensitive to alloying (~69 GPa), recycled aluminum remains a viable candidate for stiffness-critical components.

2.2.2 STRUCTURAL LOADS AND SAFETY FACTORS

Launch environments impose quasi-static accelerations (up to 15g) and random vibration loads. However, for a nanosatellite with a mass of 1.33 kg, the absolute forces are low. Finite Element Analysis (FEA) of 1U structures consistently reveals that maximum von Mises stresses rarely exceed 50 MPa under worst-case launch loads *(Napoli, 2013; Slejko et al., 2020)*.

Using Al 6061-T6 (Yield 276 MPa): Safety Factor ≈ 5.5.

Using Al 6063-T6 (Yield 214 MPa): Safety Factor ≈ 4.2.

Using Recycled Al (Yield ~145 MPa): Safety Factor ≈ 2.9. Even the lowest-grade recycled aluminum typically provides a Safety Factor nearly double the standard aerospace requirement of 1.25-1.5 *(Goddard Space Flight Center, 2018)*. This suggests that Al 6061-T6 represents a significant "over-design" for the 1U form factor.

FIGURE 2.2: Safety Margin Analysis for 1U CubeSat Under 15g Load

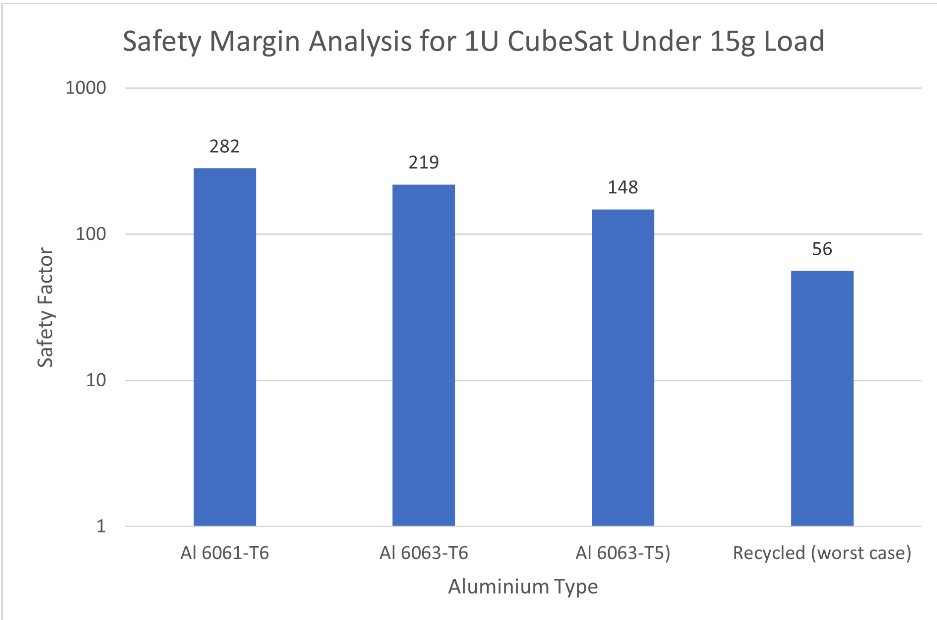


FIGURE 2.2: Safety factor comparison for 1U CubeSat rails under 15g quasi-static load. All materials, including worst-case recycled scrap, demonstrate safety factors exceeding aerospace standards (SF≥2.0) by orders of magnitude, validating the core hypothesis that CubeSat structures are vastly over-designed for static loads.

TABLE 2.2: Comparative Mechanical Properties of Aluminum Alloys

The following table synthesizes the mechanical properties of the control material against local variants to quantify the performance trade-off.

Material Designation	Source Context	Yield Strength (MPa)	Ultimate Tensile Strength (MPa)	Elastic Modulus (GPa)	Characteristics
Al 6061-T6	Aerospace Standard (Control)	276	310	68.9	High strength, structural standard.
Al 6063-T6	Lagos Architectural Market	214	241	68.9	High extrudability, lower strength.
Recycled Al (Scrap)	Local Foundry (Typical)	~145 - 160	~180	~69	High variance; prone to Fe/Si impurities.
Recycled Al (Unrefined)	Local Foundry (Cans/Mix)	55 - 120	120 - 150	~69	Low strength; suitable for non-load parts.
Al 5052-H32	Lagos Marine Market	193	228	70.3	Non-heat treatable; strain hardened.

2.3 PHYSICS OF ANISOTROPY

While aluminum forms the primary load-bearing "skeleton" (rails), the research proposes Fused Deposition Modeling (FDM) for the secondary structure (chassis/frames). Argument 3 asserts that the inherent anisotropy of FDM is a critical constraint that must be managed through design.

FDM is fundamentally distinct from isotropic manufacturing (like injection molding) due to its layer-by-layer deposition physics.

The mechanical behavior of FDM-printed carbon fiber composites is fundamentally directional. As the polymer melt is extruded through the nozzle, shear forces align both the polymer chains and the chopped carbon fibers along the print path, concentrating load-bearing capacity in the longitudinal (XY) plane. Because covalent bonds along the fiber and polymer backbone are orders of magnitude stronger than the secondary intermolecular forces governing cross-sectional cohesion, the extruded bead exhibits high tensile strength and stiffness in the XY direction. For carbon fiber reinforced Nylon 12 (PA12-CF), this in-plane tensile strength reaches 76–84 MPa, comparable to some light structural alloys *(Islam et al., 2025)*.

Strength in the build direction (Z-axis), however, is governed by an entirely different mechanism. Rather than fiber reinforcement, Z-axis cohesion depends on the thermal diffusion of polymer chains across the interlayer interface, a process described by the reptation model (*Paek et al., 2022)*, in which chains must remain above the glass transition temperature long enough to entangle with the adjacent layer. In FDM, the rapid cooling required for dimensional accuracy truncates this diffusion window, producing an interlayer bond that is effectively unreinforced polymer weakened further by interfacial porosity. Chopped fibers are too short to bridge this interface and contribute nothing to Z-axis resistance. The result is a brittle, void-rich plane that acts as a pre-existing crack under triaxial loading, a structural liability confirmed across multiple CF-reinforced thermoplastic systems *(Ramírez-Prieto et al., 2025; Islam et al., 2025)*.

2.3.1 QUANTIFYING THE WEAKNESS

The mechanical penalty of this anisotropy is severe. For carbon fiber reinforced Nylon 12 (PA12-CF), the tensile strength in the XY orientation can reach **76-84 MPa** (comparable to some plastics) *(Ramírez-Prieto et al., 2025)*, but the transverse in-plane strength (Z-axis strength) can drop to **25-40 MPa**, less than 50% of the in-plane strength *(Islam et al., 2025)*. More critically, the Z-axis ductility is negligible; failure is brittle and sudden. Under random vibration, where loads are applied tri-axially, a Z-axis layer line aligned with a bending moment acts as a pre-existing crack, leading to delamination.

2.3.2 COMPARATIVE PERFORMANCE OF CARBON FIBER THERMOPLASTIC MATRICES

Carbon fiber reinforced polylactic acid (CF-PLA) represents the most accessible entry point into FDM composite printing, benefiting from low material cost, wide filament availability, and minimal moisture sensitivity. In the XY plane, CF-PLA achieves tensile strengths in the range of 50–65 MPa, sufficient for light secondary structural members under modest loading *(Ramírez-Prieto et al., 2025)*. However, the material carries two critical disqualifying characteristics for primary structural use in a CubeSat context. First, PLA is inherently brittle; its failure mode under tensile loading is sudden and without significant plastic deformation, meaning it provides no warning before fracture under vibration loading. Second, its heat deflection temperature of approximately 55–60°C falls within the LEO thermal cycling envelope of −40°C to +80°C defined by NASA GEVS, meaning an uncontrolled thermal excursion could induce dimensional distortion or premature failure in a PLA structural component. CF-PLA is therefore retained in this project exclusively for the non-structural showpiece model, where thermal and dynamic loads are absent.

Carbon fiber reinforced Polyamide 12 (CF-Nylon 12 / PA12-CF) offers the strongest mechanical profile of the three materials evaluated. In-plane tensile strength reaches 76–84 MPa, accompanied by a higher elastic modulus and superior toughness relative to PLA-based composites, providing a more gradual failure response under cyclic loading *(Ramírez-Prieto et al., 2025; Islam et al., 2025)*. Its heat deflection temperature of approximately 100–120°C comfortably exceeds the LEO thermal ceiling, removing thermal instability as a structural risk. The primary processing challenge is hygroscopicity: PA12 absorbs between 0.25% and 1.9% moisture by weight under ambient conditions, which degrades interlayer bond strength by degrading chain mobility at the Z-interface during printing. This necessitates a mandatory pre-bake protocol, typically 80°C for 4–8 hours, before printing and before flight integration. Provided this protocol is observed, CF-Nylon 12 is the strongest, most thermally stable, and most structurally capable of the accessible FDM composites evaluated, and is selected as the primary composite material for the Structural Engineering Model.

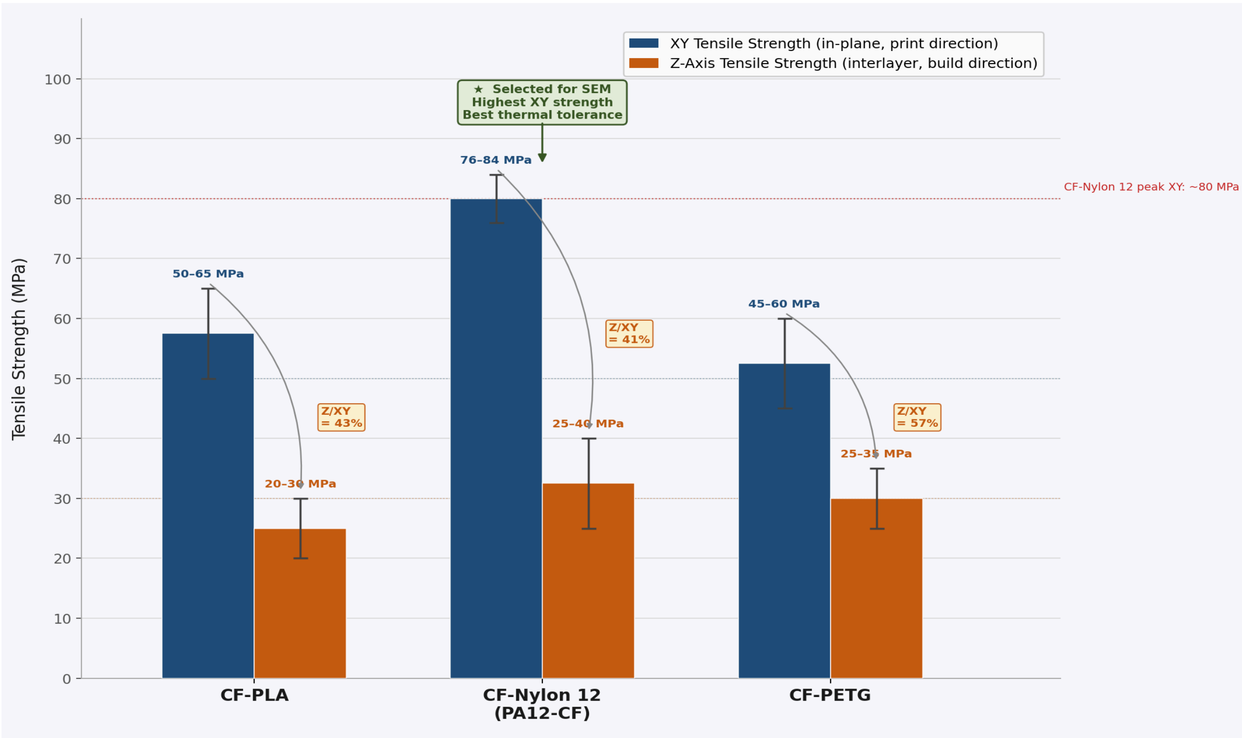
Carbon fiber reinforced PETG (CF-PETG) occupies an intermediate position between CF-PLA and CF-Nylon 12. *Economides et al. (2024)* investigated the quasi-static and viscoelastic properties of additively manufactured CF-PETG under varying infill parameters, reporting XY tensile strengths in the range of 45–60 MPa, broadly comparable to CF-PLA but achieved with meaningfully better interlayer adhesion. This improved Z-axis cohesion arises from PETG's lower cooling shrinkage relative to PLA, which reduces the thermal gradient at the interlayer interface and extends the diffusion window during deposition, producing a less voided bond zone. Scanning electron microscopy of CF-PETG fracture surfaces has confirmed interfacial bonding improvement in annealed samples, with fibre pull-out as the dominant failure mode, indicating that the matrix-interface, rather than bulk material, remains the governing failure mechanism. However, CF-PETG's heat deflection temperature of approximately 70–80°C is only marginally above the LEO thermal ceiling and its absolute XY strength is lower than CF-Nylon 12, making it a secondary option that is not selected for the SEM but is noted as a viable alternative for future work in thermally benign environments.

TABLE 2.3: Comparative Mechanical and Thermal Properties of Carbon Fiber–Reinforced FDM Thermoplastics

All values represent FDM-printed specimens at 100% infill in the XY orientation unless stated. Z-axis values reflect interlayer tensile strength (build direction). Ranges reflect variance across print parameters and filament brands reported in the literature.

Property	CF-PLA	CF-Nylon 12(PA12-CF)	CF-PETG
XY Tensile Strength (MPa)	50–65	76–84	45–60
Z-Axis Tensile Strength (MPa)	20–30	25–40	25–35
Elastic Modulus (GPa)	6–8	5–8	4–6
Density (g/cm³)	~1.24	~1.10	~1.27
Max Service Temp (°C)	~55–60	~100–120	~70–80
Hygroscopic Risk	Low	High	Low–Medium
Outgassing Risk	Medium–High	Medium (pre-bake req.)	Medium
Z-Axis Anisotropy Ratio(Z ÷ XY strength)	~40–46%	~33–48%	~42–58%
Role in This Project	Showpiece /non-structural only	Selected for the SEM primary composite structure	Intermediate option; not selected

FIGURE 2.3: Comparison of XY (in-plane) and Z-axis (interlayer) tensile strength for three carbon fiber–reinforced FDM thermoplastics considered in this study



Navy bars represent in-plane (XY) strength; orange bars represent Z-axis interlayer strength. Error bars indicate the reported literature range. The Z/XY ratio annotated above each group quantifies the anisotropy penalty; the proportion of XY strength retained in the build direction. CF-Nylon 12 achieves the highest absolute XY strength (76–84 MPa) and is selected for the Structural Engineering Model. All three materials exhibit severe Z-axis weakness (33–58% of XY strength), confirming that print orientation is a critical design variable for any FDM structural component subjected to triaxial launch loads.

Sources: Ramírez-Prieto et al. (2025), AIMS Materials Science, 12(6), 1153–1175; Islam et al. (2025), Journal of Composites Science, 9, 186.

2.3.3 DESIGN IMPLICATIONS OF FDM ANISOTROPY

The anisotropic property data presented in Table 2.3 and Figure 2.3 impose non-negotiable rules on the geometric design of any FDM structural component within the hybrid CubeSat chassis. The primary rule is print orientation: all faces that will bear tensile or bending loads during launch must be oriented so that those loads act within the XY plane, where fiber alignment provides maximum resistance. Concretely, this means that panels intended to carry bending moments induced by random vibration must be printed flat, with their largest face parallel to the build plate, so that the bending stress acts along the fiber direction rather than across layer interfaces *(Islam et al., 2025)*. The second rule governs fastener geometry: no threaded hole or heat-set insert pocket should be oriented such that the fastener's pull-out axis is aligned with the Z-direction. A screw pulling along the Z-axis loads the weakest plane of the part, the void-rich interlayer interface, in pure tension, and at Z-axis strengths of 25–40 MPa, the margin against the pull-out loads generated during random vibration at 14.1 Grms becomes critically thin *(Kastner et al., 2023)*. All fastener holes must therefore be oriented so that pull-out loads act in the XY plane, where the reinforced polymer matrix provides adequate resistance.

FIGURE 2.3.3: Print Orientation Diagram

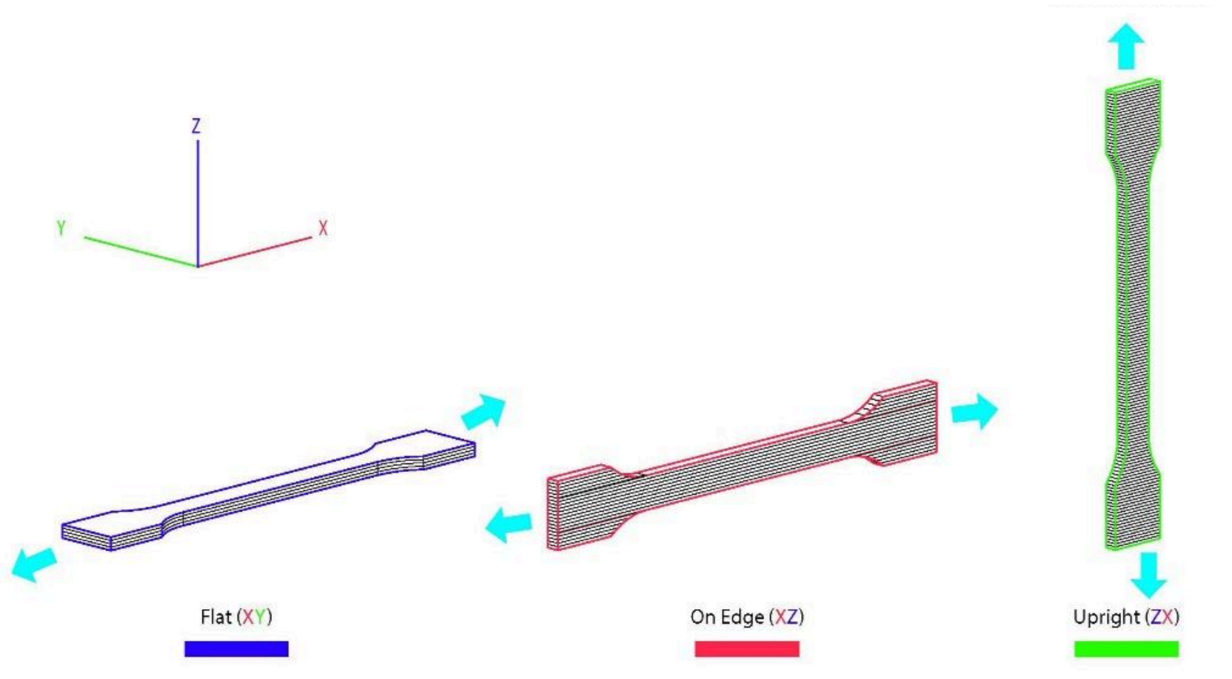


Image gotten from Orion Additive Manufacturing

The selection of CF-Nylon 12 over CF-PLA for the Structural Engineering Model is directly justified by two columns in Table 2.3. The temperature column shows CF-PLA's 55–60°C service limit falling below the 80°C upper bound of the LEO thermal environment defined by NASA GEVS, a disqualifying condition requiring no further analysis *(Slejko et al., 2020; Goddard Space Flight Center, 2018)*. The XY strength column reinforces this conclusion: CF-Nylon 12's 76–84 MPa in-plane strength provides a structural margin approximately 25–30% greater than CF-PLA's 50–65 MPa, which directly translates to a higher factor of safety under the quasi-static 15g launch acceleration loads *(Goddard Space Flight Center, 2018; Islam et al., 2025)*. While CF-Nylon 12's hygroscopic behaviour introduces a processing constraint, this is a controlled risk, mitigated by a documented pre-bake protocol, rather than an intrinsic structural limitation *(Ramírez-Prieto et al., 2025)*. CF-PLA's brittleness and thermal vulnerability, by contrast, are material properties that cannot be designed around without abandoning the polymer entirely.

2.4 HEAT-SET INSERTS

The fourth argument addresses the integration of these dissimilar materials. A hybrid structure combining rigid aluminum rails with a compliant FDM chassis is feasible only if the joining strategy accounts for the mechanical and thermal incompatibilities.

Directly threading screws into printed plastic is structurally unsound for flight hardware. The shear strength of plastic threads is low, and the viscoelastic nature of Nylon leads to "creep," causing preload loss over time *(Hogade et al., 2025)*. The proposed solution is the use of Heat-Set Inserts. These are brass nuts with knurled exteriors. They are installed by heating them with a soldering iron and pressing them into an undersized hole in the print. The heat melts the surrounding plastic, which reflows into the knurls and solidifies. This creates a composite interface where pull-out strength and torque resistance are dictated by the bulk shear strength of the plastic plug diameter, rather than the delicate threads *(Hogade et al., 2025)*. This "hybrid joint" allows high-torque stainless steel fasteners (M3) to be used reliably with printed parts.

2.4.1 THERMAL EXPANSION MISMATCH (CTE)

A critical challenge in LEO is thermal cycling (eclipse to sunlight), where temperatures can swing from -40°C to +80°C *(Goddard Space Flight Center, 2018).*

Aluminum 6063: Has a Coefficient of Thermal Expansion (CTE) of approximately **23.4** $\mu m/m \cdot K$ **(***ASM International., 1990)*

Nylon 12 CF: Exhibits anisotropic thermal expansion. In the flow direction (reinforced by fibers), the CTE is relatively low (~**37-46** $\mu m/m \cdot K$). However, in the cross-flow direction (transverse), the CTE is dominated by the polymer matrix and can exceed **100** $\mu m/m \cdot K$ *(Khina et al., 2025)*.

If the FDM chassis is rigidly bolted to the aluminum rails at both ends, this mismatch will generate significant thermal stresses. Since the FDM part expands more than the aluminum, it will be put into compression (potentially buckling) while putting the rails in tension. To mitigate this, "Frugal Innovation" demands sophisticated design: using slotted holes or flexure-based mounting points to allow the chassis to "float" relative to the rails, decoupling thermal growth from structural loads.

2.5 THE HIGH-END BIAS

While CubeSats and 3D printing are well-researched domains, a specific gap exists at the intersection of *unverified local materials* and *consumer-grade additive manufacturing*.

Current literature on "3D Printed CubeSats" predominantly focuses on high-performance thermoplastics like PEEK or ULTEM 9085, processed on industrial systems (e.g., Stratasys Fortus). While these materials are flight-qualified, industrial FDM systems capable of processing PEEK require build temperatures exceeding 350°C and cost an order of magnitude more than consumer-grade printers *(Paek et al., 2022)*. Conversely, "Low-Cost CubeSat" literature often assumes the use of standard COTS aluminum frames (Pumpkin/ISISpace), ignoring the supply chain barriers discussed in Argument 1.

2.5.1 THE MISSING METHODOLOGY

There is a lack of rigorous qualification data for a hybrid system combining **Architectural Grade Al 6063** (with its "inferior" yield strength) and **Prosumer-Grade Nylon-CF** (printed on consumer-grade FDM printers costing under $1,000 USD). The specific research gap is the validation of this material pair under GEVS conditions.

1. *Can Al 6063 rails maintain the P-POD dimensional tolerances (8.5 ± 0.1 mm) after random vibration?*
2. *Will the Z-axis of a consumer-printed Nylon frame delaminate under the 14.1 Grms acoustic loads?*
3. *How does the variance in Nigerian recycled aluminum affect the statistical safety factors?*

This project addresses this gap by creating a "Material Atlas" and validating the structural integrity of this specific, accessible material combination.

TABLE 2.5: Research Gap Analysis Matrix

Research Domain	**Current State of Art**	**Limitations for Emerging Space Nations**	**Addressed by Current Project?**
CubeSat Structures	Monolithic Al 6061/7075 Machined Frames.	High cost, import dependence, rigid supply chain.	**Yes:** Validates accessible Al 6063/Scrap.
AM in Space	PEEK/ULTEM printing; Sintered Metal (DMLS).	Requires industrial printers (>$100k); expensive materials.	**Yes:** Validates accessible Nylon-CF/PLA-CF.
Material Characterization	High-fidelity data for virgin aerospace alloys.	No data for locally sourced Nigerian aluminum in space contexts.	**Yes:** Characterizes local scrap & extrusions.
Hybrid Integration	Aerospace bonding; complex riveted assemblies.	Requires specialized tooling/processes.	**Yes:** Develops accessible Insert-based joining.

2.6 FRUGAL INNOVATION

The final argument posits that "Frugal Innovation" is not about accepting lower safety standards, but about replacing *cost* with *characterization*.

Frugal innovation, as theorised by *Meagher (2017)*, refers to the deliberate redesign of products, processes, and supply chains to deliver adequate, rather than optimal, performance at significantly reduced cost, primarily by substituting expensive imported inputs with locally available alternatives and simplifying designs to match the capabilities of local manufacturing infrastructure. In development economics, the concept emerged from observations of grassroots technological adaptation in the Global South, where resource constraints force engineers and entrepreneurs to find unconventional solutions that established industries overlook. *Meagher (2017)* acknowledges that this approach can generate genuine technological capability within emerging economies when it builds on existing local skills and material ecosystems rather than simply imitating lower-cost versions of foreign products. However, Meagher also raises a significant critique: that frugal innovation in practice can cannibalise formal manufacturing sectors, suppress wages, and entrench informality rather than building the institutional foundations needed for sustained industrial development; a concern particularly relevant to the West African context where informal aluminium processing dominates the local supply chain. However, this critique applies primarily to consumer product markets; for high-stakes engineering applications, frugal innovation is validated only through rigorous testing rather than cost-cutting at the expense of safety *(Prakash & Bhattacharya, 2025)*.

Frugal engineering, as championed by agencies like ISRO (India), focuses on maximizing utility while minimizing resource consumption *(Prakash & Bhattacharya, 2025)*. For a University-class CubeSat mission in LEO (lifetime < 1 year), the 10-year fatigue life and radiation shielding properties of Al 6061 are arguably superfluous. Al 6063 is "good enough" *if and only if* its properties are rigorously characterized and the design is adjusted to respect its limits (e.g., thicker wall sections to compensate for lower yield strength). This validity is contingent on testing. The project does not merely "hope" the materials hold; it mandates testing the "Structural Engineering Model" (SEM) to GEVS acceptance levels to prove survivability.

TABLE 2.6: Literature Gap Analysis

Focus Area	**Dominant Literature Narrative**	**Identified Gap**	**Relevance to Project**
Material Selection	Dominance of Al 6061/7075 and Titanium.	Limited research on "Architectural" alloys (6063) in flight structures.	Validates Al 6063 as a viable structural candidate.
Composite Structures	Continuous fiber composites or PEEK.	Scarcity of data on short-fiber FDM (Nylon-CF) for primary structure.	Assesses FDM-CF for secondary structural roles.
Manufacturing	High-precision CNC & Industrial AM.	Lack of "Design for Manufacturing" (DFM) for limited resource labs.	Proposes DFM for local FabLabs/Foundries.
Qualification	Full Space-Grade Qualification.	Lack of "risk-tolerant" qualification protocols for educational CubeSats.	Proposes tailored validation for LEO missions.

The six arguments presented in this review collectively establish the technical and economic foundation for the proposed hybrid structural approach. The global CubeSat standard mandates aerospace-grade aluminum, based on a supply chain assumption that is structurally inaccessible to emerging space economies, such as Nigeria, creating a material standard disconnect that is institutional rather than purely technical. Locally available Aluminum 6063 and recycled scrap are metallurgically inferior to Al 6061-T6, yet the absolute structural loads imposed on a 1U CubeSat are sufficiently low that both local alloys provide safety factors exceeding the aerospace minimum of 1.25–1.5, confirming that the performance gap is relative rather than disqualifying. FDM carbon fiber composites introduce severe Z-axis anisotropy that precludes direct substitution for aluminum structural members, but this constraint is designable, print orientation rules and fastener geometry decisions can systematically route critical loads into the reinforced XY plane. Heat-set insert joining resolves the mechanical and thermal incompatibility at the aluminum-to-polymer interface, providing pull-out resistance and torque retention that direct threading in thermoplastics cannot reliably sustain under vibration loading. The precedent set by ISRO's frugal engineering philosophy confirms that cost-constrained material choices are legitimate in high-stakes applications, provided they are backed by rigorous characterisation and empirical validation rather than assumption. Taken together, these five technical arguments are made viable by the sixth: that frugal innovation, properly executed, is not an acceptance of reduced safety margins but a redefinition of where engineering effort is invested, from procurement to verification.

No existing study has produced a qualification dataset for a hybrid CubeSat primary structure combining locally sourced Aluminum 6063 or recycled scrap with consumer-grade FDM carbon fiber composites, validated against NASA GEVS random vibration acceptance levels in a developing-country fabrication context. This is the precise gap this research addresses.

The "weakness" is relative, not absolute. Similarly, the anisotropy of FDM Carbon Fiber composites is a manageable constraint, provided that design principles, specifically optimal print orientation and insert-based joining, are rigorously applied to mitigate Z-axis weakness and thermal expansion mismatch.

By addressing the identified research gaps, specifically the lack of a qualification dataset for this low-cost material combination, this project validates a "Frugal Engineering" pathway.

CHAPTER THREE

METHODOLOGY

The research adopts a mixed-methods approach, combining experimental material science with computational mechanics and physical prototyping.

TABLE 3.1: Research Design Matrix

The methodology is structured into five sequential phases:

Phase	**Activity**	**Method/Tool**	**Output**
I	Material Characterization	Tensile Testing (ASTM E8/D638)	Yield Strength
II	Structural Design	CAD (SolidWorks), DFM	Hybrid Assembly Model
III	Simulation	FEA (ANSYS/SolidWorks Sim)	Random Vibration Response (3σ stress)
IV	Fabrication	CNC Machining, FDM Printing	Structural Engineering Model (SEM)

V	Testing & Validation	Vibration Analysis (Proxy/Physical)	Natural Frequency (fn), Static Deflection, Dimensional Compliance; pass/fail against GEVS acceptance levels.

3.2 PHASE I: MATERIAL CHARACTERIZATION

This phase establishes the "Ground Truth" for the locally sourced materials.

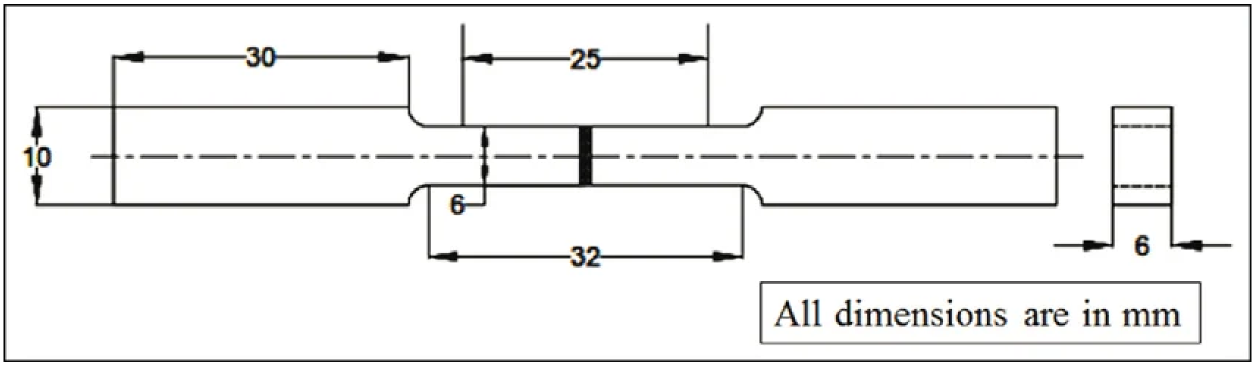
1. Sourcing:

A. **Aluminum A (Control):** Standard datasheet values *(ASM International., 1990)*
B. **Aluminum B (Local Extrusion):** Al 6063 box profiles or bars procured from local aluminum extrusion vendors in Lagos.
C. **Aluminum C (Scrap):** Cast aluminum billets sourced from Lagos foundries processing automotive scrap.
D. **Composite Filament:** Carbon Fiber reinforced Nylon (PA12-CF), PETG-CF, and PLA-CF from local 3D printing filament suppliers or online retailers serving Nigeria.

2. Specimen Preparation:

A. **Metal:** "Dogbone" specimens will be machined from the bulk aluminum samples according to **ASTM E8** (Standard Test Methods for Tension Testing of Metallic Materials) *(ASTM International., 2022).*

FIGURE 3.2.1: Tensile Test specimen (ASTM E8)

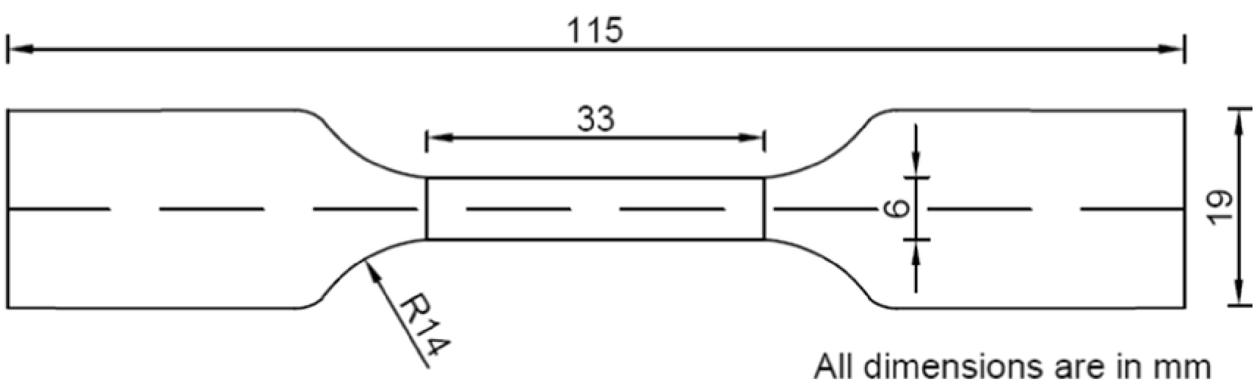


B. **Composite:** Specimens will be printed according to **ASTM D638** Type IV (Standard Test Method for Tensile Properties of Plastics) *(ASTM International., 2022)*.

   a. *Variable:* Print Orientation. Specimens will be printed in XY (Flat), XZ (On-Edge), and ZX (Upright) orientations to quantify the anisotropy ratio

   b. *Settings:* 100% Infill, 0.2mm layer height, 250-260°C nozzle temp (for Nylon) *(Ramírez-Prieto et al., 2025)*.

FIGURE 3.2.2: Tensile Test specimen (ASTM D638)



Sample Size and Statistical Approach:

1. Each material configuration will be tested with n=5 specimens minimum (per ASTM recommendations for materials with unknown variance)
2. Total test matrix:

   a. Al 6063-T6: 5 specimens (XY orientation)
   b. Recycled Al scrap (Batch A): 5 specimens
   c. Recycled Al scrap (Batch B, if variance is high): 5 specimens
   d. Nylon-CF (XY orientation): 5 specimens
   e. Nylon-CF (XZ orientation): 5 specimens
   f. Nylon-CF (ZX orientation): 5 specimens
   g. PLA-CF (XY orientation, for showpiece validation): 3 specimens
   h. Total: ~33 tensile tests
3. Results will be reported as mean ± standard deviation
4. The 5th percentile value (A-basis) will be used for FEA material properties to ensure conservative design (NASA., 2023)

3. Testing Procedure:

A. Equipment: Universal Testing Machine (Instron) available at the University of Lagos or the Standards Organisation of Nigeria (SON) laboratory.
B. Parameters: Strain rate of 5 mm/min.
C. Data Collection: Load vs. Extension data will be converted to Stress vs. Strain curves to determine the 0.2% Offset Yield Strength and Young's Modulus.

3.3 PHASE II: STRUCTURAL DESIGN AND OPTIMIZATION

1. Architecture:

The design utilizes a "Hybrid Skeleton" approach to optimize the strengths of each material:

1. **Primary Load Path (Rails):** Four longitudinal rails machined from the best-performing Local Aluminum (6063 or Scrap). These rails carry the compressive launch loads from the P-POD deployment spring. The rails will require surface treatment to prevent cold welding in the P-POD mechanism per CDS requirements. Three options will be investigated:
   a. Hard Anodization (Type III): Industry standard but may require outsourcing to

specialized facilities in Lagos or Abuja industrial zones

b. Electroless Nickel Plating: Available from marine/automotive finishing shops in Lagos (e.g., Apapa industrial area); provides comparable hardness and galling resistance
c. Chromate Conversion Coating (Alodine): Simpler chemical process; lower performance but may be adequate for ground demonstration If none are accessible within budget, this SEM will use uncoated rails for structural testing only, with the understanding that flight hardware requires proper surface treatment.

Hard anodization is the preferred option; electroless nickel plating is the secondary option if Type III anodization facilities cannot be located within the project budget; chromate conversion coating is the tertiary option for the SEM only

2. **Secondary Structure (Chassis):** Top/Bottom plates and internal bulkheads printed in **Carbon Fiber Nylon**. These parts provide lateral stiffness (shear panels) and mounting points for the PCB stack (PC/104 standard).
3. **Load Transfer:** The rails act as columns. The printed chassis prevents the rails from buckling.

2. Joining Strategy:

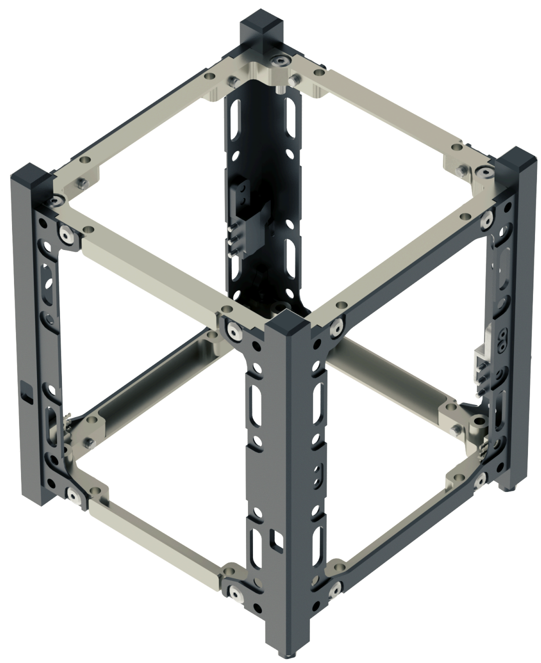
1. The rails will feature counterbored holes.
2. The printed parts will feature heat-set threaded insert pockets (Hogade et al., 2025).
3. M3 Stainless Steel screws will secure the rails to the chassis. This avoids tapping the aluminum (which might be soft/porous in the case of scrap) or the plastic directly.

Direct tapping of recycled scrap aluminum is avoided due to the risk of tap breakage in porous or hard-phase-rich material.

3. Mass Budget:

The design will target a structural mass of < 300g, leaving ~1 kg for the payload. The lower density of the CF-Nylon (1.2 g/cm³) compared to Aluminum (2.7 g/cm³) contributes to this efficiency *(Ramírez-Prieto et al., 2025)*.

FIGURE 3.3: Hybrid Chassis Architecture Diagram (SAMPLE)



3.4 PHASE III: FINITE ELEMENT ANALYSIS AND STRUCTURAL SIMULATION

All finite element analyses are conducted in **SolidWorks Simulation Premium**, selected for its native SolidWorks CAD integration, availability within the University of Lagos FabLab, and validated static and frequency-domain solvers. Three sequential studies are conducted: quasi-static stress analysis, modal analysis, and random vibration response analysis.

3.4.1 GEOMETRY AND MATERIAL PROPERTY ASSIGNMENT

The Phase II hybrid chassis assembly is imported into SolidWorks Simulation as a multi-body assembly, with aluminum rail bodies and FDM panel bodies assigned as separate components. Material properties are assigned manually from Phase I **A-basis (5th percentile)** experimental values — not from the SolidWorks default library — ensuring simulation results reflect the conservative lower bound of measured performance (SAE International, 2012).

FDM panels are modelled as **orthotropic** materials. The Z-axis elastic modulus (typically 30–50% of XY) is applied to the panel thickness direction to represent interlayer compliance under bending loads (Islam et al., 2025). Property inputs are summarised in Table 8.4.1.

TABLE 3.4.1: FEA material property inputs. Strength and modulus values populated from Phase I A-basis results.

Property	**Al 6063-T6(Local)**	**Recycled AlScrap**	**CF-Nylon 12(XY)**	**CF-Nylon 12(Z)**
Yield Strength (MPa)	Phase IA-basis	Phase IA-basis	N/A	N/A
UTS (MPa)	Phase IA-basis	Phase IA-basis	Phase IA-basis	Phase IA-basis
Elastic Modulus (GPa)	Phase IA-basis	Phase IA-basis	Phase IA-basis	Phase IA-basis
Poisson's Ratio	0.33	0.33	0.35	0.35
Density (g/cm³)	2.70	Phase Itesting	1.10	1.10
Material Model	Isotropic	Isotropic	Orthotropic	Orthotropic

3.4.2 MESH GENERATION AND CONVERGENCE

A **second-order (parabolic) tetrahedral mesh** is applied to all bodies. Second-order elements are used over first-order elements for superior stress gradient capture at joint interfaces, fastener holes, and insert pockets.

Convergence is established by refining global element size from 3.0 mm to 0.5 mm until successive steps produce **less than 5% change** in both peak von Mises stress and first natural frequency. Local refinements are applied at all stress concentration features.

3.4.3 STUDY 1: QUASI-STATIC STRESS ANALYSIS

Fixtures

The four P-POD rail contact faces are assigned fixed constraints per CDS Rev. 14.1 *(The CubeSat Program, Cal Poly SLO, 2022)*.

Loading

1. **Axial (Z-axis):** 15g (147.15 m/s^2) body force in the launch direction (NASA GEVS GSFC-STD-7000A, 2018).
2. **Lateral (X-axis):** 7.5g body force (50% of axial load, per standard aerospace practice). Mass dummy bodies for OBC, EPS, and battery stack are included to correctly distribute inertial loads.

Output

Von Mises stress distributions extracted for all bodies. Factor of Safety calculated as:

FoS = A-basis Yield Strength ÷ Maximum von Mises Stress

3.4.4 STUDY 2: MODAL ANALYSIS

Boundary conditions are identical to Study 1. The modal solver extracts the first ten undamped natural frequencies and mode shapes. The primary success criterion is $\mathbf{fn_1 > 100\ Hz}$ *(The CubeSat Program, Cal Poly SLO, 2022)*. Mode shapes are inspected to identify whether the governing mode is panel bending, rail torsion, or a localised printed component mode, informing any geometry optimisation before fabrication.

3.4.5 STUDY 3: RANDOM VIBRATION ANALYSIS

The analysis uses **modal superposition**, building on Study 2 mode shapes. The PSD input is taken from **NASA GEVS Table 2-III**, applied as base excitation at the rail contact faces (14.1 Grms, 20–2000 Hz, NASA Goddard Space Flight Center, 2018). A structural damping ratio of $\boldsymbol{\zeta = 0.02}$ is applied — conservative relative to typical assembled CubeSat damping, meaning dynamic amplitudes are over-predicted.

The solver produces RMS stress distributions. **3σ stress** (three times RMS, per aerospace convention) is extracted at critical nodes and compared against the A-basis yield strength. Success criterion: **3σ von Mises stress < 70% of A-basis yield** at all structural nodes.

3.4.6 SUCCESS CRITERIA AND VALIDATION

The FEA phase passes only if all five criteria below are simultaneously satisfied. Any failure triggers geometry modification and a repeat of the simulation cycle before Phase IV fabrication begins.

TABLE 3.4.2: FEA success criteria. TBD entries populated upon completion of simulation studies.

Criterion	**Metric**	**Threshold**	**Study**	**Pass?**
Structural stiffness	fn_1 (first natural freq.)	> 100 Hz	Modal	TBD
Static strength	FoS under 15g	≥ 2.0	Quasi-static	TBD
Dynamic strength	3σ von Mises / yield	< 70%	Random vib.	TBD
Z-axis composite	3σ Z-stress / Z-UTS	< 50%	Random vib.	TBD
Rail deflection	Max δ under 15g	< 0.1 mm	Quasi-static	TBD

Model validation is performed in Phase V by comparing the experimentally measured fn_1 against the Study 2 prediction. A correlation within **±15%** is accepted as adequate for an undergraduate structural study. If exceeded, orthotropic panel moduli are updated with Phase I measured values and the simulation is re-run.

3.5 PHASE IV: FABRICATION OF THE ENGINEERING MODEL

1. **CNC Machining:**
    a. The aluminum rails will be machined at the **Unilag FabLab** or a partner workshop in Lagos.
    b. The CAD files will be converted to G-Code. Special attention will be paid to the tolerance of the rail width (8.5mm +/- 0.1mm) to ensure P-POD fit.
2. **3D Printing (Additive Manufacturing):**
    a. **Printer:** FDM printer with hardened steel nozzle (available at Unilag FabLab).
    b. **Nozzle:** 0.4mm or 0.6mm Hardened Steel Nozzle.
    c. **Filament:** Nylon-CF (PA12-CF).

d. **Process:**
    i. Parts will be dried before printing (Nylon is hygroscopic) *(Ramírez-Prieto et al., 2025)*.
    ii. Orientation: Parts will be oriented to minimize Z-axis tension loads.
    iii. Enclosure: Printing will occur in a passive enclosure to prevent warping.

3. **Integration:**
    a. Installation of brass threaded inserts using a soldering iron set to ~250°C (Hogade et al., 2025).
    b. Assembly of rails and panels.
    c. Integration of "Mass Dummies" (Aluminum blocks) to simulate the battery and OBC mass, bringing the total QM mass to 1.33 kg *(The CubeSat Program, Cal Poly SLO., 2022)*.

3.6 PHASE V: TESTING AND FEASIBILITY ASSESSMENT

1. Physical Fit Check:

1. Use of a "Test Pod" or a vernier caliper to verify the dimensions against the CDS static envelope.

2. Structural Testing Protocol:

The testing approach will be tiered based on equipment availability:

Primary Method (If Accessible): Modal Testing

1. Partner with industrial facilities in Lagos (oil & gas equipment testing labs, automotive QC departments) or NASRDA to access a vibration shaker or modal exciter
2. Conduct sine-sweep testing (20-2000 Hz at low amplitude) to experimentally identify the first natural frequency
3. Compare measured frequency with FEA predictions to validate the computational model
4. *Goal*: Prove fn > 100 Hz experimentally

Secondary Method (If Shaker Unavailable): Impulse Response Testing

1. Mount the SEM on a rigid fixture
2. Apply controlled impulse using an instrumented impact hammer
3. Measure response using an accelerometer or a smartphone-based accelerometer with FFT analysis capability.
4. Perform FFT analysis to extract modal frequencies
5. While less comprehensive than shaker testing, this validates FEA-predicted natural frequencies

Tertiary Method (Fallback): Static Load Testing Only

1. Apply calibrated static loads (15g equivalent) using dead weights and fixtures
2. Measure deflections using dial indicators or digital calipers
3. Verify structure remains in elastic region (no permanent deformation)
4. *Limitation*: Does not validate dynamic performance or fatigue resistance

What Will NOT Be Attempted:

1. Drop testing is inappropriate as a vibration proxy (impact shock ≠ sustained random vibration)
2. Any method claiming to "simulate" random vibration without proper equipment is scientifically invalid

The final report will clearly state which testing method was used and acknowledge the limitations accordingly. All proxy testing methods used in place of shaker testing will be documented with explicit statements of their limitations and the corresponding reduction in confidence level for the feasibility conclusion.

3.7 EXPECTED OUTPUTS AND DELIVERABLES

The project will generate the following tangible and intangible outputs:

3.7.1 TECHNICAL DATA PACKAGE

1. **Material Atlas:** A documented dataset of the mechanical properties (Yield, UTS, E) for Lagos-sourced Aluminum 6063 and Recycled Scrap. This will be a valuable resource for future student projects in Nigeria.

The Material Atlas will be presented as a structured data table in Appendix A of the final report, documenting mean, standard deviation, and 5th percentile (A-basis) values for yield strength, UTS, and elastic modulus for each material and orientation tested, formatted for direct reuse by future student projects at the Lagos State University and other Nigerian institutions undertaking CubeSat development.

2. **Simulation Report:** A detailed FEA report showing the mode shapes, stress distribution, and safety margins of the hybrid design under GEVS loads.

3.7.2 THE STRUCTURAL ENGINEERING MODEL (SEM)

1. A physical, 1U CubeSat chassis constructed from Local Al 6063 and Nylon-CF.
2. This hardware serves as proof that a hybrid design can meet the mass and dimensional requirements of the CDS.

3.7.3 DESIGN GUIDELINES

1. A set of "Best Practices" for integrating threaded inserts into 3D printed space structures.
2. Recommendations for machining local scrap aluminum to minimize the impact of porosity.

These guidelines will be compiled as Appendix B of the final report, structured as a numbered best-practice checklist with supporting rationale drawn from the experimental and simulation results of this project; a separate standalone publication is outside the scope of this undergraduate project and is recommended as future work.

3.8 PROJECT TIMELINE

TABLE 3.8: PROJECT TIMELINE

Month	**Activity**
1	Literature Review & Sourcing of Local Al 6063/Scrap & Filaments.
2	Specimen Fabrication (Machining & Printing) & Tensile Testing (ASTM E8/D638). Analysis of Material Data. Design of Hybrid Structure (CAD).
3	Finite Element Analysis (FEA) - Modal & Random Vib. Optimization.
4	Fabrication of Engineering Model (CNC & FDM). Assembly.
5	Feasibility Testing (Vibration/Shock) & Showpiece Printing.
6	Data Synthesis, Final Report Compilation, Review, and (if time permits) display model fabrication

3.9 ESTIMATED BUDGET (PROVISIONAL)

TABLE 3.9: ESTIMATED BUDGET

Item	**Description**	**Estimated Cost (₦)**
Materials - Metal	Al 6063 Extrusion / Scrap (5kg)	₦45,000
Materials - Filament	Nylon-CF (1kg), PETG-CF (1kg) & PLA-CF (1kg)	₦200,000
Hardware	M3 Screws, Threaded Inserts, brass hardware, fastener variety pack	₦25,000
Machining	CNC Time	₦150,000
Material Testing **Surface Treatment** **Vibration Testing**	Universal Testing Machine access (12-15 specimens) Anodization outsourcing OR electroless nickel plating Shaker access, impact testing equipment, or accelerometer rental	₦100,000 ₦60,000 ₦80,000

Travel & Logistics	Materials sourcing trips, sample transport, vendor meetings	₦70,000
Documentation	Printing, binding, and photo documentation	₦20,000
Contingency (20%)	For design iterations, failed prints, and unexpected material costs	₦144,000
Total		**₦894,000**

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